

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S8 (R,S) / S8 (PT) (R) Examination May 2024 (2019 Scheme)

Course Code: EET426**Course Name: SPECIAL ELECTRIC MACHINES****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

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| 1 | List any three applications of BLDC motors. | (3) |
| 2 | Discuss the features of permanent magnets used in PMDC motors. | (3) |
| 3 | A three phase three stack variable reluctance stepper motor has 10 rotor teeth. Find its step angle. | (3) |
| 4 | What are the modes of operation of stepper motors? | (3) |
| 5 | Draw and explain the phasor diagram of Synchronous Reluctance Motors. | (3) |
| 6 | What are the noise mitigation techniques used in SRM? | (3) |
| 7 | List any three applications of servomotors. | (3) |
| 8 | Draw and explain the Torque - speed characteristics of an armature-controlled DC servomotor with variation in control voltage. | (3) |
| 9 | Compare torque speed characteristics of single-phase induction motor and hysteresis motor. | (3) |
| 10 | Discuss the classification of linear motors. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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| 11 | a) Explain the constructional details of a BLDC motor | (7) |
| | b) Explain any one control scheme of BLDC motor with diagram. | (7) |

OR

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| 12 | a) Explain the principle of operation of PMSM. | (7) |
| | b) With a neat block diagram, explain the self-control scheme of PMSM. | (7) |

Module II

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| 13 | a) With the aid of necessary diagrams and truth table explain the operation of a permanent magnet stepper motor in the half step mode. | (8) |
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- b) Explain the construction of a hybrid stepper motor. (6)

OR

- 14 a) With the help of necessary diagrams, Explain the working of a VR stepper motor. (8)
- b) Sketch and explain static and dynamic characteristics of stepper motors. (6)

Module III

- 15 a) Derive the torque equation of a synchronous reluctance motor. (7)
- b) Explain any two power converter circuits used in switched reluctance motors. (7)

OR

- 16 a) Draw and explain the torque-speed characteristics of a switched reluctance motor. Give any two applications of SRM. (6)
- b) With necessary diagram, explain the principle of operation of a 10/4 switched reluctance motor. (8)

Module IV

- 17 a) Derive the transfer function of a field-controlled DC Servomotor and draw the block diagram. (8)
- b) Explain the principle of operation of an AC servomotor. (6)

OR

- 18 a) Explain the constructional details of DC servomotor. (9)
- b) Explain the differences that a servomotor possesses in comparison to conventional industrial motors. (5)

Module V

- 19 a) Explain the construction and working of Linear Reluctance Motors. (10)
- b) Explain the transverse edge and end effects in LIM. (4)

OR

- 20 a) Explain with diagram the principle of operation of a Repulsion Motor? Also give its any two applications (8)
- b) Describe the principle of operation of Universal motor. (6)
